[redacted]

1

$$G(s) = \frac{s^2+8s+12}{s^3+9s^2+23s+15}$$

a) 3. Ordnung

b)

$$V(s)(s^3+9s^2+23s+15) = U(s)(s^2+8s+12)$$

$$\dddot{v}(t) + 9\ddot{v}(t) + 23\dot{v}(t) + 15v(t) = \ddot{u}(t) + 8\dot{u}(t) + 12u(t)$$

c) ja es ist ein LTI System, weiles zeitinvariant und linear ist.

d)

$$g(t) = \mathcal{L}^{-1}\{G(s)\}$$

$$\frac{s^2+8s+12}{(s+5)(s+1)(s+3)} = \frac{A}{(s+5)} + \frac{B}{(s+1)} + \frac{C}{(s+3)}$$

$$s^2+8s+12 = A\cdot(s+1)(s+3) + B(s+5)(s+3) + C(s+5)(s+1)$$

$$s^2+8s+12 = A(s^2+3s+s+3) + B(s^2+3s+5s+15) + C(s^2+s+5s+5)$$

$$s^2+8s+12 = A(s^2+4s+3) + B(s^2+8s+15) + C(s^2+6s+5)$$

s^2: $A+B+C=1$

s^1: $4A+8B+6C=8$

s^0: $3A+15B+5C=12$

mit TR

$A = -\frac{3}{8}$

$B = \frac{5}{8}$

$C = \frac{3}{4}$

$$G(s) = \frac{-\frac{3}{8}}{(s+5)} + \frac{\frac{5}{8}}{(s+1)} + \frac{\frac{3}{4}}{(s+3)}$$

$$g(t) = -\frac{3}{8}e^{-5t} + \frac{5}{8}e^{-t} + \frac{3}{4}e^{-3t} \quad \text{für } t>0$$

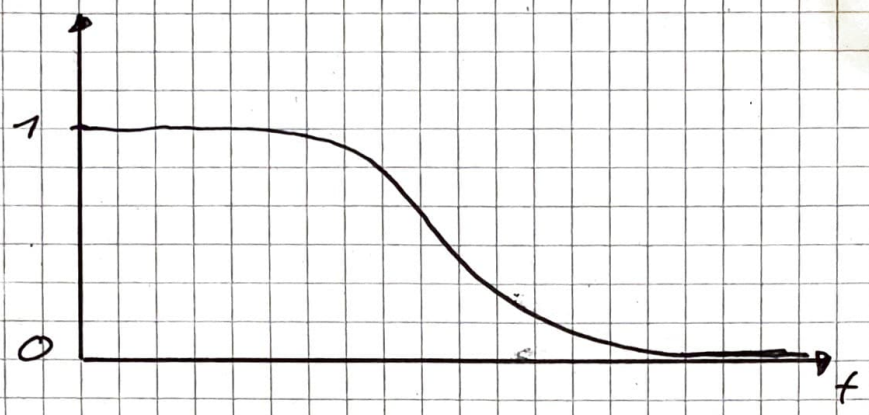
e) $\lim_{t\to 0} g(t) = \cancel{\lim_{t\to\infty} s\cdot G(s) = s\cdot\frac{s^2+8s+12}{s^3+9s^2+23s+15}}$

$\lim_{t\to 0} g(t) = -\frac{3}{8} + \frac{5}{8} + \frac{3}{4} = 1 \Rightarrow$ Anfangswert

7

[redacted]

$$\lim_{t\to\infty} g(t) = \lim_{s\to 0} s\,G(s) = \lim_{s\to 0} s\,\frac{s^2+8s+12}{s^3+9s^2+23s+15} = 0 \Rightarrow \text{Endwert}$$

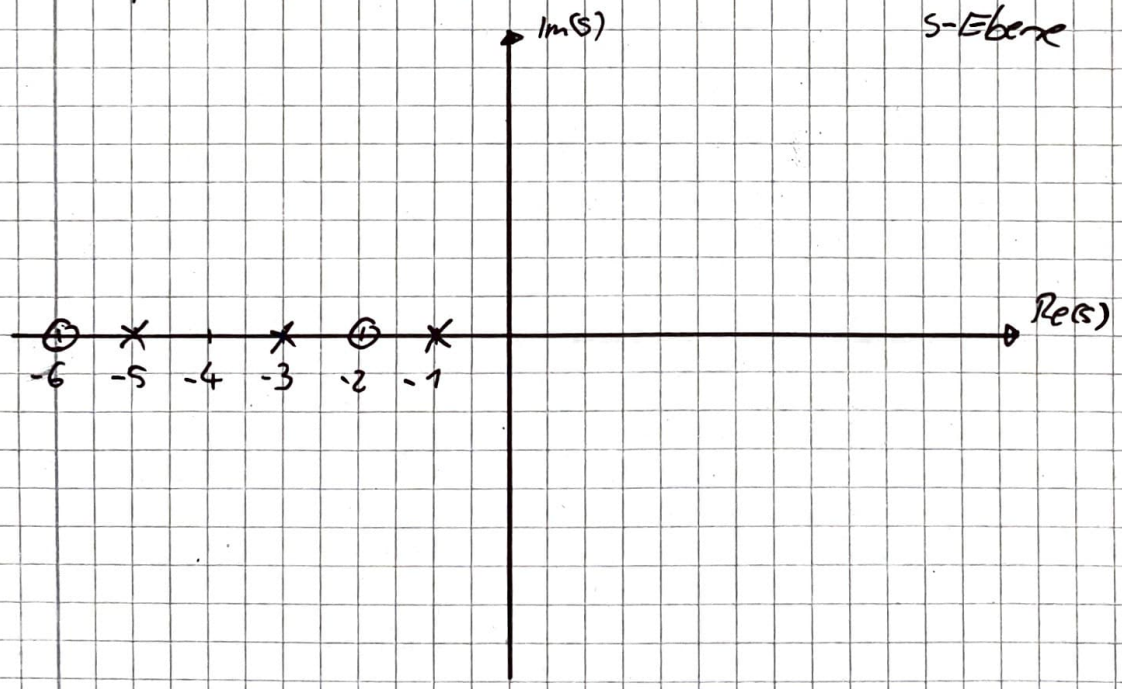


f)

$(s+5)(s+1)(s+3) = 0$

g) nst: $s_1 = -5$ $s_2 = -3$ $s_3 = -1$

polst: $s_4 = -2$ $s_5 = -6$



h) nein, alle polstellen und Nullstellen befinden sich links von der Imaginären Achse

i) ja alle pole sind links von der Imaginären Achse

[redacted]

1

j)

2

a) PT_3 - Regelstrecke

PI[redacted] - Regler

b) $G_0(s) = G_R(s) \cdot G_S(s) = \frac{k_R}{s \cdot T_N}(1+sT_N) \cdot \frac{400}{(4+s)(1+s \cdot 0{,}00[redacted]s)(1+s)}$

$= \frac{k_R \cdot (1+sT_N)\, 400}{s \cdot T_N\, (4+s)(1+s \cdot 0{,}00[redacted]s)(1+s)}$

c) $k_R = 1 \qquad T_N = 1$

$G_0(s) = \frac{400}{s\,(4+s)(1+s \cdot 0{,}00[redacted]s)} = \frac{400}{s\,(1+\frac{1}{4}s)(1+s\,0{,}00[redacted]s)}$

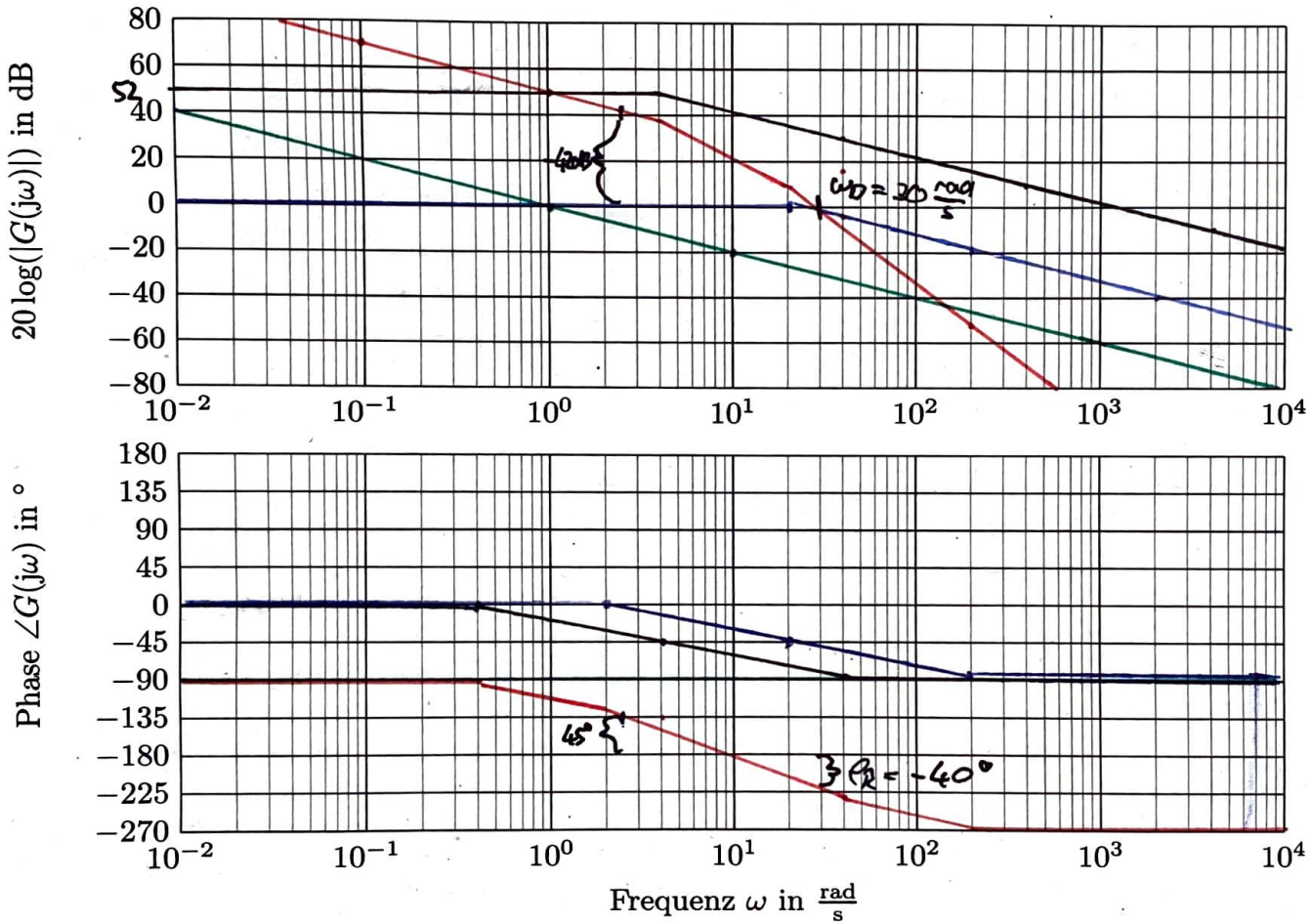
$G_0(s) = \frac{1}{s} \cdot \frac{400}{(1+\frac{1}{4}s)} \cdot \frac{1}{(1+s\,0{,}00[redacted]s)}$

d) $\omega_D = 30\,\frac{rad}{s} \qquad \varphi_R = -40°$

e)

[redacted]

(2) $G_0(s) = \frac{1}{s} \frac{400}{(1+\frac{1}{4}s)} \left(\frac{1}{1+0{,}05}\right)$



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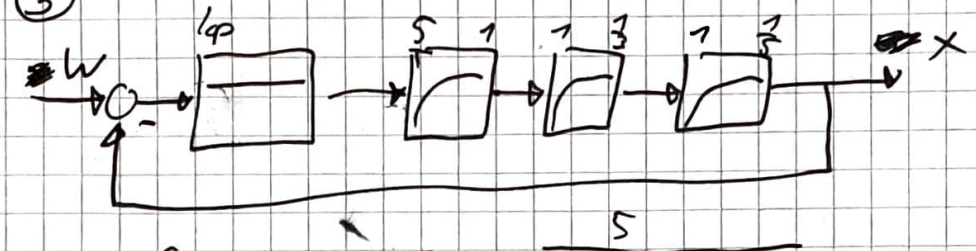
③

a)

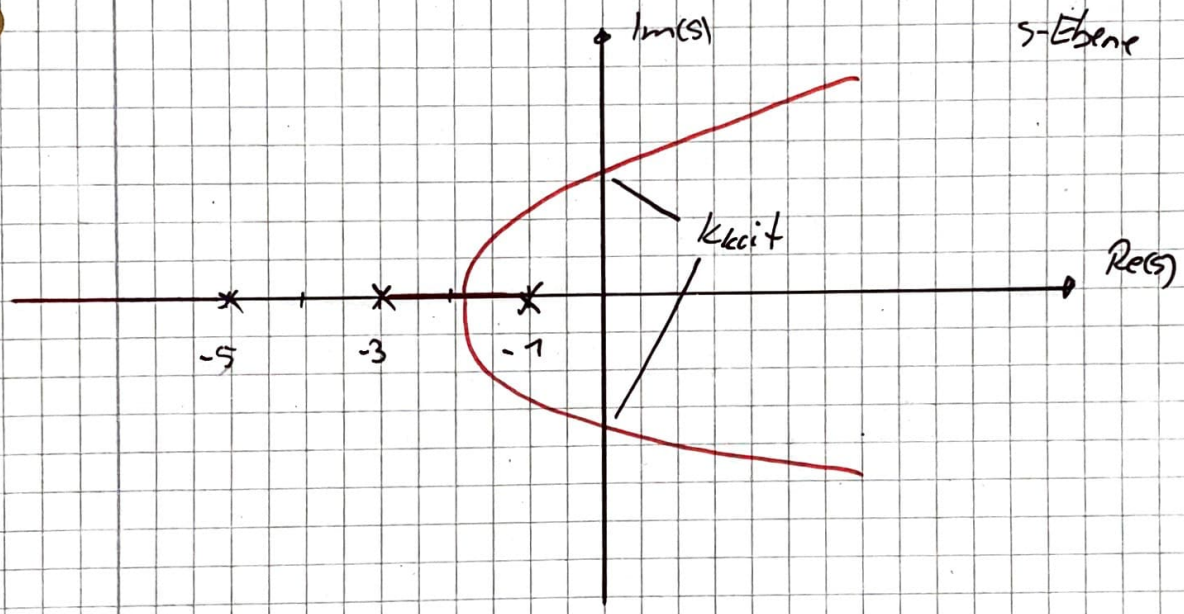


$G_R(s) = k_p \qquad G_S(s) = \dfrac{5}{(s+1)(s+3)(s+5)}$

b)

$$G_W(s) = \frac{k_p 5}{(s+1)(s+3)(s+5) + k_p 5} \qquad G_0(s) = \frac{k_p 5}{(s+1)(s+3)(s+5)}$$

c) $s_1 = -1 \quad s_2 = -3 \quad s_3 = -5$



d)

$$N(s) = (s^2 + 4s + 3)(s+5) + k_p 5$$
$$= s^3 + 4s^2 + 3s + 5s^2 + 20s + 15 + k_p 5$$
$$= s^3 + 9s^2 + 23s + 15 + k_p 5$$

$$N_W(s) = s^3 + \sigma_3 s^2 + \omega_1^2 s + \omega_1^2 \sigma_3$$

s^2: $9 = \sigma_3$

s^1: $23 = \omega_1^2$

s^0: $15 + k_p 5 = \omega_1^2 \sigma_3 \Rightarrow k_p = \dfrac{\omega_1^2 \sigma_3 - 15}{5} = \dfrac{23 \cdot 9 - 15}{5} = 38{,}4 = k_{p\,krit}$

[redacted]

③

e) $k_{pa} = \overset{0,6}{\cancel{0,5}} \cdot k_{pkrit} = \overset{0,6}{\cancel{0,5}} \cdot 38,4 = 23,04$

$\cancel{T_{Na} = 0,5 \cdot k_{pkrit} = 0,5 \cdot 38,4 = 19,2}$

$\cancel{T_{Va} = 0,12}$

ω_{krit} = Imaginärteil von polstelle bei k_{pkrit} (mit TR)

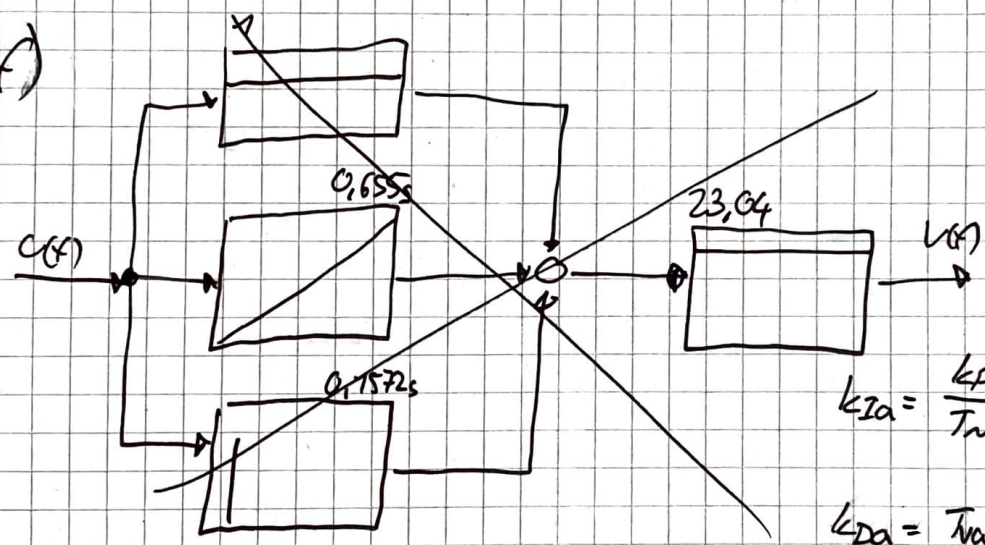
$\omega_{krit} = 4,8 \frac{rad}{s}$

$T_{krit} = \frac{2\pi}{\omega_{krit}} = \frac{2\pi}{4,8} = 1,31$

$T_{Na} = 0,5 \cdot T_{krit} = 0,5 \cdot 1,31 = 0,655s$

$T_{Va} = 0,12\, T_{krit} = 0,1572s$

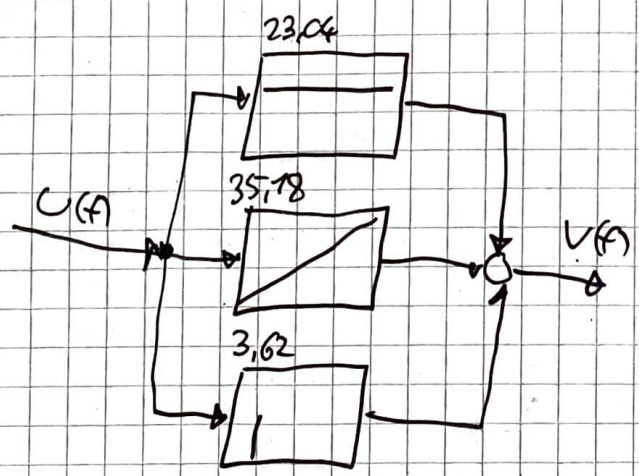
f)



$k_{Ia} = \frac{k_{pa}}{T_{Na}} = \frac{23,04}{0,655} = 35,18$

$k_{Da} = T_{Va} \cdot k_{pa} = 0,1572s \cdot 23,04$

$= 3,622$



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(2)

f) $\varphi_R = 45°$ bei $\omega = 2{,}5 \frac{rad}{s}$

Amplitudenreserve = 42 dB

$-42\,dB = 20 \log k_p\,dB$

$k_p = 10^{\frac{-42}{20}} = 7{,}9 \cdot 10^{-3}$

g) ja

(5)

a) 4 Äste

[redacted]

④

a) $F_A - k_w \cdot v(t)^2 = m \cdot \dot{v}(t) \Rightarrow k_w \cdot v(t)^2 + m \cdot \dot{v}(t) = F_A$

b) Ap $v = v_0$

$$\frac{d}{dv}\left(k_w v(t)^2\right)\Big|_{v_0} = 2 k_w v_0$$

$$2 k_w v_0 \Delta v + m \Delta \dot{v} = \Delta F_A$$

c) $$G_{(s)}(s) = \frac{\Delta V(s)}{\Delta F_A(s)}$$

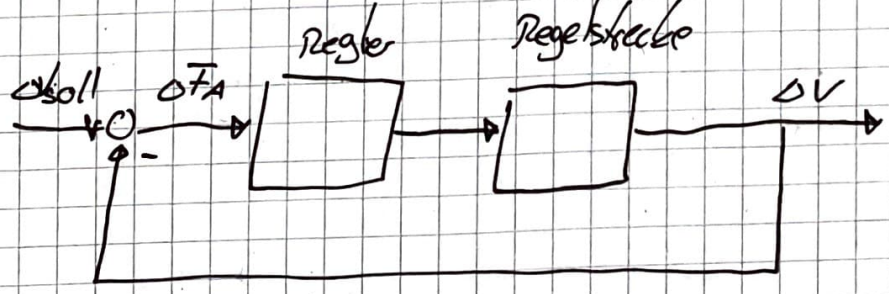
$$2 k_w v_0 \cdot \Delta V(s) + m \cdot s \cdot \Delta V(s) = \Delta F_A(s) \cdot 1$$

$$\Delta V(s)\,(2 k_w v_0 + m \cdot s) = \Delta F_A(s) \cdot 1$$

$$G(s) = \frac{\Delta V(s)}{\Delta F_A(s)} = \frac{1}{(2 k_w v_0 + m s)} = \frac{1}{2 k_w v_0 \left(1 + \frac{m}{2 k_w v_0} s\right)}$$

PT_1 verhalten $\quad T = \frac{m}{2 k_w v_0} \quad k_p = 1$

d)



e) PI-Regler um die Polstelle zu kompensieren → Stabilität
I-Anteil für stat. Genauigkeit